Malware Analysis Report

Red Team Assessment – File Detection & Behavioral Analysis

🔹 1. Executive Summary:

This assessment evaluates the effectiveness of modern security controls in detecting known malicious signatures using controlled test artifacts. An EICAR-standard file was generated and analyzed across multiple platforms, including VirusTotal and Hybrid Analysis. Results confirm strong signature-based detection while demonstrating expected minimal behavioral activity in sandboxed environments.

🔹 2. Objective:

The objective of this exercise was to:

- Validate antivirus detection capabilities using a standardized test file
- Assess multi-engine threat intelligence correlation via VirusTotal
- Perform controlled sandbox execution and behavioral inspection
- Identify indicators of compromise (IOCs) associated with the analyzed file

🔹 3. Methodology:

🔸 3.1 Test File Creation:

A safe simulation of malware was conducted using the EICAR test string, which is globally recognized for validating antivirus detection mechanisms without introducing real threats.

```
echo X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H* > test.eicar
```

The file was successfully generated within the test environment and verified for analysis.

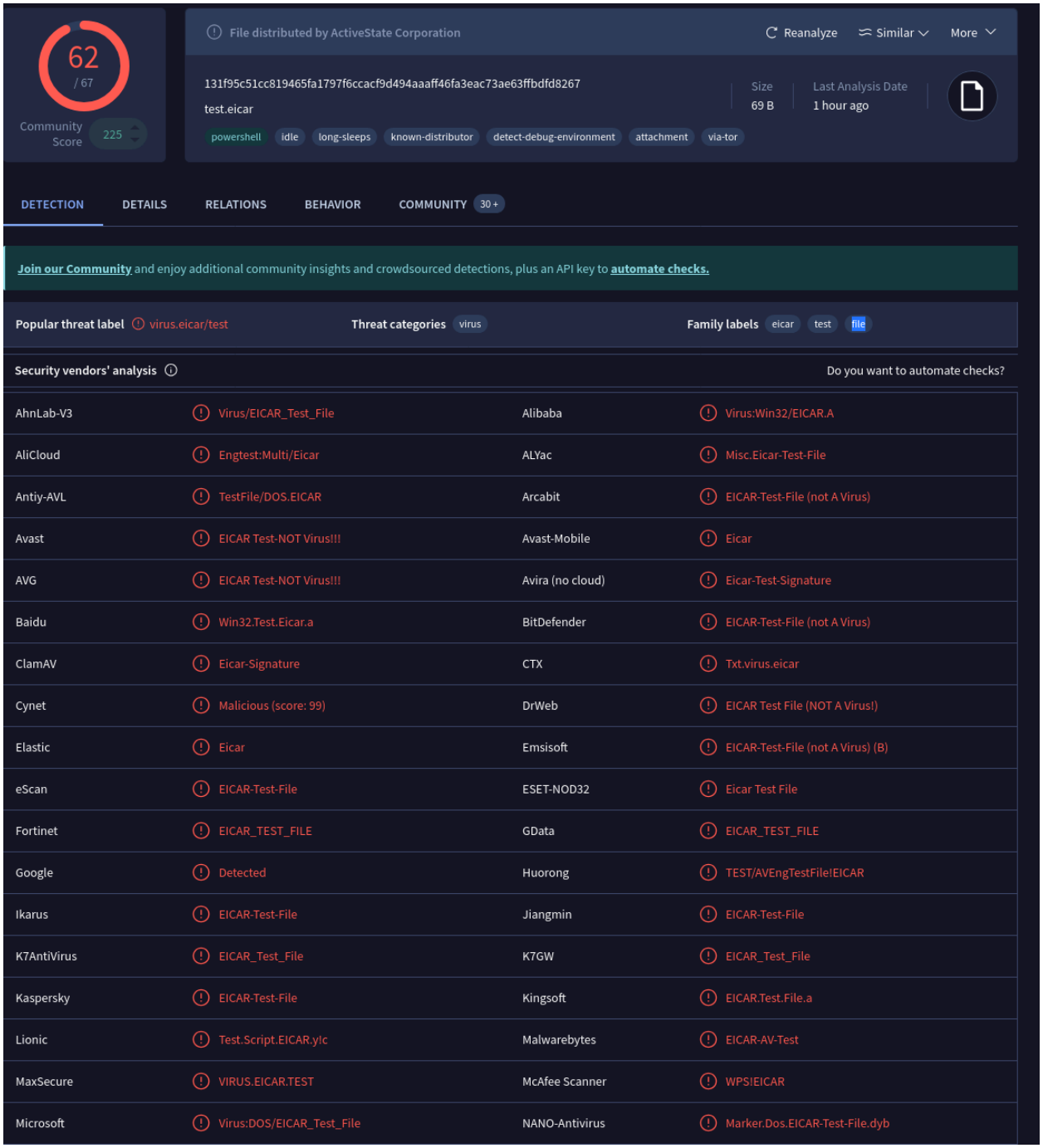
🔸 3.2 Static Analysis – VirusTotal

The test file was uploaded to VirusTotal for static analysis across multiple antivirus engines.

- The platform generated unique cryptographic hashes (MD5, SHA256), ensuring file traceability

- Multiple security vendors flagged the file as malicious
- Detection signatures consistently identified the file as **EICAR-Test-File**
- The detection ratio confirmed high confidence in signature-based identification

This phase validated that industry-standard antivirus solutions effectively recognize known malicious patterns.



🔸 3.3 Dynamic Analysis – Hybrid Analysis Sandbox

The file was submitted to a controlled sandbox environment for behavioral execution and runtime inspection.

- The sandbox assigned a threat score based on known signatures
- Execution behavior was minimal, as expected for a non-functional test payload
- No persistence mechanisms, network communication, or system modifications were observed
- The file was flagged based on signature recognition rather than active malicious behavior

HYBRID ANALYSIS | Sandbox | Quick Scans | File Collections | Resources | Request Info

Analysis Overview

Request Report Deletion | Show Sample Content

Submission name:	131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
Size:	69B
Type:	com executable
Mime:	text/plain
SHA256:	131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
Submitted At:	2018-04-06 19:51:13 (UTC)
Last Anti-Virus Scan:	2026-03-13 12:29:51 (UTC)
Last Sandbox Report:	2026-03-13 12:29:50 (UTC)

malicious

Threat Score: 100/100
AV Detection: 69%
Labeled As: EICAR
#tag

Post | Link | E-Mail

0 Community Score 0

Anti-Virus Results

Updated 4 days ago - Click to Refresh

MetaDefender
Multi Scan Analysis

Malicious (18/26)

More Details

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Falcon Sandbox Reports (142)

Characteristics Legend | Show All As List | Submit

Not all reports are visible. 137 error reports are hidden.

Show All As List

Windows 11 64 bit
eicar.com
May 15th 2025 10:30:49 (UTC)
Malicious
Threat Score: 100/100
Labeled As: -
Indicators: 2
Characteristics:

Windows 7 32 bit
eicar.com
March 19th 2025 20:35:58 (UTC)
Malicious
Threat Score: 100/100
Labeled As: -
Indicators: 3 18
Characteristics:

Windows 7 64 bit
EICAR.com
January 2nd 2025 11:21:52 (UTC)
Malicious
Threat Score: 100/100
Labeled As: -
Indicators: 2
Characteristics:

Windows 7 64 bit
EICAR.com
January 2nd 2025 11:21:52 (UTC)
Malicious

Windows 10 64 bit
eicar.com
October 25th 2024 23:30:36 (UTC)
Malicious

🔹 4. Indicators of Compromise (IOCs):

Indicator Type	Value
File Name	test.eicar
File Type	Text File
Detection Signature	EICAR-Test-File
Hash (SHA256)	Generated via VirusTotal
Classification	Known Test Malware

🔹 5. Analysis Summary (50 Words):

The EICAR test file was successfully detected by multiple antivirus engines on VirusTotal, confirming effective signature-based detection. Hybrid Analysis sandbox assigned a threat score but showed no significant behavior. This demonstrates that security tools reliably identify known threats while accurately distinguishing non-functional test files from active malware.

🔹 6. Key Findings:

- Strong detection consistency across multiple antivirus engines
- Accurate classification of known test malware signatures
- No behavioral anomalies observed during sandbox execution
- Effective correlation between static and dynamic analysis platforms

7. Recommendations:

- Continue leveraging multi-engine scanning platforms for threat validation
- Integrate sandbox analysis into incident response workflows
- Regularly test detection capabilities using safe simulation artifacts like EICAR
- Complement signature-based detection with behavioral and heuristic analysis

🔹 7. Conclusion:

This assessment confirms the reliability of modern security solutions in detecting known malicious signatures through signature-based mechanisms. The absence of behavioral activity

reinforces the controlled nature of the EICAR file while demonstrating the importance of combining static and dynamic analysis for comprehensive threat evaluation.